

## HDOC Day-14 — Menu-Driven Program Using Switch-Case

### Problem Statement

Write a menu-driven C program using **switch-case** to input a positive integer and perform one of the following operations:

1. Check whether all digits of the number are unique.

### Algorithm

1. Start.
2. Input a positive integer num.
3. Initialize an array freq[10] with all values as 0.
4. Store num in temp.
5. Repeat while temp > 0:
  - Find the last digit: digit = temp % 10.
  - Check freq[digit].
  - If it is already 1, the number has a repeated digit.
  - Otherwise, set freq[digit] = 1.
  - Remove the last digit: temp = temp / 10.
6. If a repeated digit is found, display "**Not Unique**".
7. Otherwise, display "**Unique**".
8. Stop.

### Test Case Table

| Test Case | Input  | Expected Output | Actual Output | Result |
|-----------|--------|-----------------|---------------|--------|
| 1         | 12345  | Unique          | Unique        | Pass   |
| 2         | 12234  | Not Unique      | Not Unique    | Pass   |
| 3         | 987654 | Unique          | Unique        | Pass   |
| 4         | 112233 | Not Unique      | Not Unique    | Pass   |
| 5         | 56789  | Unique          | Unique        | Pass   |

### C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int num, temp, digit, i;
```

```
    int freq[10] = {0};
```

```
    int unique = 1;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &num);
```

```
    temp = num;
```

```
    while (temp > 0)
```

```
    {
```

```
        digit = temp % 10;
```

```
        if (freq[digit] == 1)
```

```
        {
```

```
            unique = 0;
```

```
            break;
```

```
        }
```

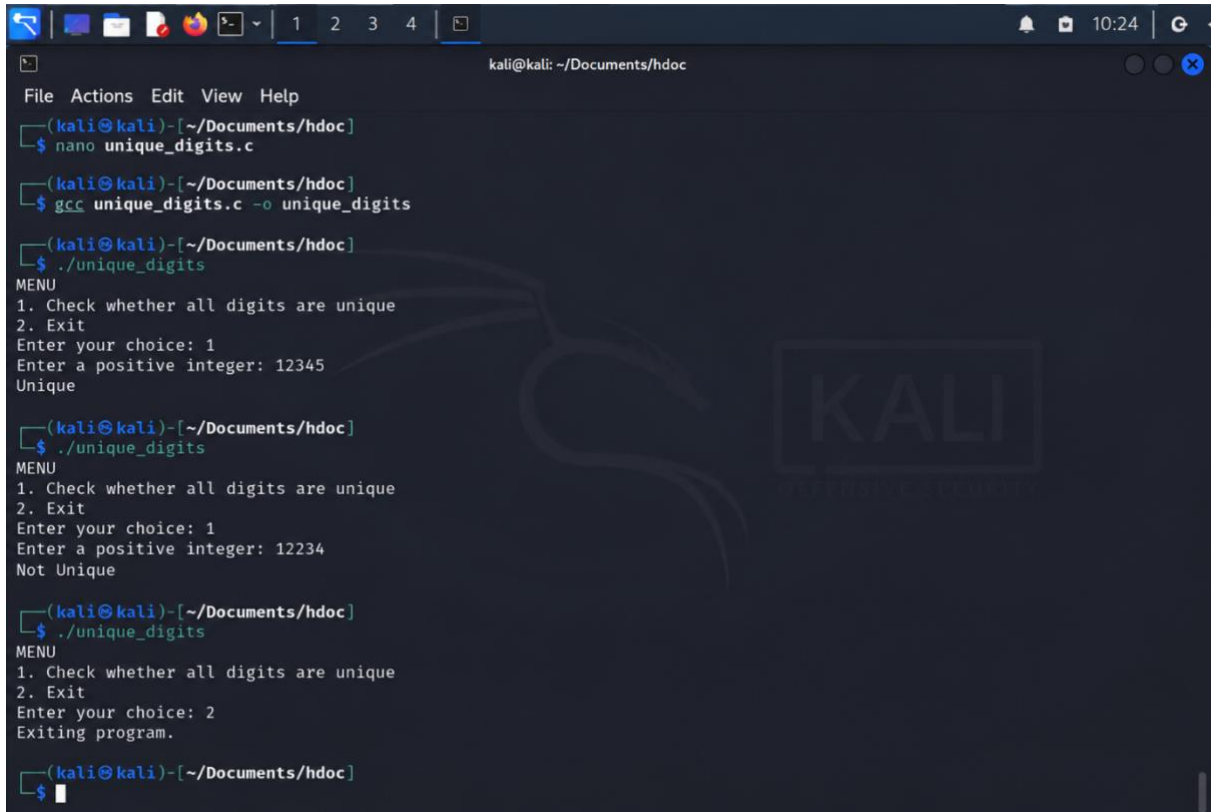
```
        freq[digit] = 1;
```

```
        temp = temp / 10;
```

```
    }
```

```
    if (unique == 1)
```

```
    printf("Unique\n");  
  
else  
  
    printf("Not Unique\n");  
  
return 0;  
}
```



```
kali@kali: ~/Documents/hdoc  
File Actions Edit View Help  
(kali@kali)-[~/Documents/hdoc]  
$ nano unique_digits.c  
(kali@kali)-[~/Documents/hdoc]  
$ gcc unique_digits.c -o unique_digits  
(kali@kali)-[~/Documents/hdoc]  
$ ./unique_digits  
MENU  
1. Check whether all digits are unique  
2. Exit  
Enter your choice: 1  
Enter a positive integer: 12345  
Unique  
(kali@kali)-[~/Documents/hdoc]  
$ ./unique_digits  
MENU  
1. Check whether all digits are unique  
2. Exit  
Enter your choice: 1  
Enter a positive integer: 12234  
Not Unique  
(kali@kali)-[~/Documents/hdoc]  
$ ./unique_digits  
MENU  
1. Check whether all digits are unique  
2. Exit  
Enter your choice: 2  
Exiting program.  
(kali@kali)-[~/Documents/hdoc]  
$
```

**Result:** The program successfully checks whether all digits of the entered positive integer are unique using a switch case menu